

Top performers and top journals: Persistent concentration in scientific publishing

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Abstract

In this research, we analyze the relationship between publishing productivity and access to highly prestigious journals, treating publishing in top journals as a stratification mechanism selecting publishing elites. We study $N = 144,314$ Polish scientists publishing for 30 years (1992–2021) and their $Nart = 433,546$ unique research articles published in the period. Using bibliometric data from Scopus, we compare the scientists belonging to the top productivity decile (the upper 10%, termed top performers) and the remaining population of scientists (90%) by discipline and period (five six-year periods). We measure the share of publications in prestigious segments of journals, with particular reference to the 90th–99th percentiles, and we use nonlinear journal prestige-normalized productivity. Our results indicate that access to top journals (defined as the top 10% of journals indexed in Scopus) is powerfully and permanently concentrated in the group of top performers in all disciplines and periods studied. The differences between top performers and the other scientists are primarily of a qualitative nature: they are seen almost exclusively at the top of the journal hierarchy rather than in its bottom or middle segments. Our logistic regression models indicate the complementarity of quantity and quality: publishing intensity increases the probability of membership in the elite segment of top performers, especially when it is coupled with publishing in prestigious journals. Our results suggest that top journals function as selection gates to academic careers and that they function as durable mechanisms of elite reproduction in science.

1. Introduction

Inequalities in publishing productivity have been documented for about a century (Cole & Cole, 1973; de Solla Price, 1963; Lotka, 1926; Merton 1968). The distribution of publications in all disciplines is extremely steep: a small number of scientists account for a disproportionately high part of all knowledge production (Allison & Stewart, 1974; Kwiek, 2016). Sociological analyses of stratification in science have shown that the concentration in science does not refer only to publication quantity but also to citations, access to resources, and recognition and awards (Cole & Cole, 1973). A key element in explaining the permanence of these inequalities over time is cumulative advantage: in a system of rewards and communication in science, earlier successes increase the probability of later successes, and recognition is distributed in an asymmetric and self-reinforcing manner (Allison & Stewart, 1974; Allison et al., 1982). In other words, success (in science) breeds success, and the rich (in science) get richer.

But productivity concentration is not the only mechanism that feeds academic stratification. What is equally important – and is often critical to successful academic careers – is the concentration of

scholarly journals through which knowledge is published because journal prestige functions as an institutional signal of quality and visibility (Heckman & Moktan, 2020; Leahey, 2007). From the perspective of citation behavior, the impact is related not only to article quality: citations too reflect an author's reputation, their national and international visibility, and the prestige of publication channels. Citations, in a word, are sensitive to journal hierarchies (Bornmann & Daniel, 2008).

In the last two decades, publishing in highly prestigious journals (historically, always important in science as a provider of international recognition) has become a central reference point in many research assessment systems. Expectations of publishing in top journals are standard among national research funding agencies (Lyytinen et al., 2007), and they are an absolute must for the European Research Council (Rodríguez-Navarro & Brito, 2019). In most extreme examples, this logic takes the form of career gatekeeping through a narrow set of elite journals. In economics, for instance, as an extensive analysis by Heckman and Moktan shows, the location of a publication in the so-called “top-five” journals is powerfully linked to positive tenure and promotion decisions (Heckman & Moktan, 2020). Many disciplines have their narrowly defined lists of top journals in which the vast majority of scientists are never able to publish or in which they publish just once (as the first author has shown for top journals in higher education research, about 80% of all authors in the six top published only one paper in the 20 years studied, and only 3.3% had five publications or more; Kwiek, 2021). Different disciplines have different publishing cultures that define what is regarded as a top journal.

In biology, medicine, chemistry, and physics, the system of top journals is global, relatively stable, and based on Impact Factor (from WoS) or CiteScore (from Scopus) measures; in the social sciences, hierarchies of journals exist, but they are more contestable, with economics being the most hierarchized discipline; and in the humanities, the hierarchies of journals are more dispersed, less formalized, and less unequivocal (Hammarfelt, 2017). In technical and engineering fields, hierarchies of journals are often linked to hierarchies of conferences. Finally, in applied and interdisciplinary fields, there are different competitive lists of top journals (e.g., FT50 in management). For biology and medicine, top journals include *Nature* and *Science*; for economics, the top five include *The American Economic Review* and *Econometrica* (Heckman & Moktan, 2000); for management, top journals include *Management Review* and *Administrative Science Quarterly*; and for higher education research, they include *Higher Education* and *Studies in Higher Education*.

Not all institutions are involved to the same extent in the prestige game based on publication in top journals (Kwiek, 2021). Elite institutions in national systems tend to focus more on top journals compared with average institutions: what matters for research-intensive universities may not matter at all for comprehensive teaching-focused universities. There are also substantial cross-national differences in the role of publications in top journals for hiring, promotion, and research funding decisions (see discussions of top journals in different systems in, e.g., Bak & Kim, 2019; Fochler et al., 2016; Lindahl, 2018; Mouritzen & Opstrup, 2020; Shibayama & Baba, 2015; Sutherland, 2018).

Previous studies have often used a simplified separation of productivity (publications) and impact (citations), with productivity defined as the number of publications per scientist (within a unit of time). However, as Abramo and D'Angelo (2014) showed, such an account of productivity is conceptually problematic and leads to misguided assessments of research efficiency at various levels. They propose quality-adjusted measures of productivity that link publishing output to quality and co-authorship (Abramo & D'Angelo, 2014). In practice, this means that in studies of scientific elites, it is not enough to ask, “Who publishes more?” “Who publishes where?” also needs to be asked, and the impact that selective access to prestige channels has on positions in the science

system must also be assessed. In this research, we use nonlinear journal prestige-normalized productivity (see explanations in section 3.3. Methods).

The aim of this article is to empirically disentangle two dimensions of stratification in science: publishing productivity and access to the most prestigious journals. Moving beyond simple quantitative measures, we investigate whether the advantage of top performers (the upper 10% of scientists in publishing productivity) over the remaining scientists (90%) is linear and dispersed in the entire hierarchy of journals – or whether it emerges selectively and only in the highest segments of productivity. We pay special attention to the segment of journals from the 90–99 percentile (the upper 10% of journals) in the Scopus dataset – which, in STEM disciplines, is a powerfully selective segment of the global publishing circuit.

We link our analysis of publication elites to classical analyses of inequalities and cumulative advantage in science (Cole & Cole, 1973; de Solla Price, 1963; Lotka, 1926; Merton, 1968). We show that the prestige of the publication channel is a critical dimension of visibility and selection in science (Bornmann & Daniel, 2008; Leahey, 2007). We view the patterns discovered in the context of works that treat top journals as powerfully supporting careers, especially visible in disciplines with strong hierarchies of journals (as in economics; Heckman & Moktan, 2020). We change the productivity perspective from the question of “who publishes more” (quantity) to the question of “who publishes in what channels” (quantity and *ex ante* quality, *ex post* quality being related to citations), and we discuss the consequences of this new perspective for stratification in science.

In empirical terms, we studied 144,314 unique scientists publishing over 30 years (between 1992 and 2021) and their 433,546 unique research articles published in that period and indexed in Scopus. The data were aggregated into five six-year analytical periods and 15 STEM disciplines.

2. Literature review

The first theoretical pillar of this research comprises studies on top performers in science, which grow out of a long tradition of analyses showing that publication productivity is structurally unequal (Abramo et al., 2009; Abramo et al., 2013; Abramo et al., 2017; Ioannidis et al., 2014; Rosen, 1981; Xie, 2014). Classical works concluded that the distribution of productivity at the individual level is extremely steep: the majority of scientists publishes little or nothing, and a small minority produces a considerable share of knowledge (de Solla Price, 1963; Lotka, 1926). As Lotka (1926, p. 317) remarked in a paper presenting his “inverse square law” of scientific productivity, “it would be of interest to determine, if possible, the part which men of different calibre contribute to the progress of science.” Classical observations about the role of the highly productive tiny minority of scientists in knowledge production laid the foundations for later interpretations of social stratification in science in which knowledge production is hierarchically organized, and recognition and resources (of various types, from funding and top positions in universities to research infrastructure) are allocated unevenly (Cole & Cole, 1973).

In the sociology of science, of special importance is an approach in which inequalities in science result from institutional and cumulative mechanisms. Merton (1968, p. 7) applied the idea of cumulative advantage not only to individuals but also to institutions: “the rich get richer at a rate that makes the poor become relatively poorer. Thus, centers of demonstrated scientific excellence are allocated far larger resources for investigation than centers which have yet to make their mark.” Cole and Cole (1973) described science as a stratification system in which prestige, visibility, and access to resources form a feedback loop with productivity. In their account, better positions in the

scientific field do not result exclusively from individual differences in talent or effort – mechanisms of selection and the reproduction of hierarchies are involved. These interpretations were supported by, e.g., Allison and Stewart (1974), who showed that differences in productivity can be partly explained by the process of cumulative advantage; however, the accumulative advantage hypothesis is not able to explain productivity differences between the youngest scientists (where the “sacred spark hypothesis” could work: some scientist are not good – they are just superb, from the very beginning; see Stephan, 2012; Stephan & Levin, 1996). Allison and Stewart (1974) argue that cumulative advantage mechanisms should produce inequalities as cohorts of scientists grow older. Allison et al. (1982) developed these logics into a more general theory of inequalities in science explaining that advantages in recognition and resources increase in a self-reinforcing way. Instead of synthetic cohorts, they used true cohorts of scientists. They also suggested that their theory of inequality in science might be generalizable to other areas, e.g., to the distribution of wealth.

The principal theoretical framework for cumulative advantage processes is the Matthew effect, which means that earlier recognition and success increase the probability of later recognition and success (both symbolic and institutional) and that both tend to be concentrated: “eminent scientists get disproportionately great credit for their contributions to science while relatively unknown scientists tend to get disproportionately little credit for comparable contributions” (Merton, 1968, p. 2). Although Merton’s focus was on allocating credit and recognition (Merton, 1973), in bibliometric analyses that followed, the Matthew effect was interpreted more widely: as a mechanism that acts through institution prestige, the reputation of the author, and recognition of communication channels, including scholarly journals.

Classical literature provides two arguments important to our research questions. First, top performers (also termed “eminent scientists,” “star scientists,” or “prolific scientists”) are a structural element of the science system (Abramo et al., 2009; Fox & Nikivincze, 2021; Prpić, 1996). And second, inequalities in productivity and recognition are supported by cumulative mechanisms, which may be coupled with selective access to prestigious publication channels.

The second theoretical pillar important to our research is the set of studies on measures of prestige in science. In academic practice, citations and journal prestige often function as proxy indicators of quality. But the literature shows that citations are multidimensional measures encompassing quality, visibility, and reputation. Bornmann and Daniel (2008) indicate that citations are not a simple reflection of research quality: they also depend on journal prestige, author reputations, and the structure of the field.

Journal prestige is a critical conversion mechanism (Latour & Woolgar, 1986): publications in prestigious journals become more visible, they have a higher citation probability, and they offer faster diffusion in the science community, all of which, in the long run, strengthen the author’s position in the field (Bak & Kim, 2019; Fochler et al., 2016; Lindahl, 2018). From the perspective of academic careers, visibility functions as a resource that can be exchanged for awards and opportunities. Leahey (2007) shows that visibility and specialization are significant determinants of academic salaries and that productivity influences careers, to a large extent, through mechanisms of visibility.

If journal prestige increases visibility and strengthens cumulative advantage, then access to top journals may act as an independent mechanism of stratification – not only as a reward for productivity but also as an element of elite selection and reproduction. If “quantitative” productivity (publication counts) is only partly correlated with “qualitative” productivity (publications linked to journals in which they are published), then there is a question of whether publishing elites should be

defined through sheer publication numbers or publications in specific (prestigious) journals. We are especially interested in top journals as a potential area where “qualitative productivity” is produced and where cumulative advantage may operate most strongly.

This article draws from our previous research. We have shown elsewhere that the role of top performers in national knowledge production is relatively equal across 11 European systems. Top performers are responsible for about 50% of all publications in those systems – which suggests a structural rather than accidental character of European productivity elites (Kwiek, 2016). Using large-scale survey data, we have developed a catalog of characteristics of top performers (long working hours, long research hours, research orientation rather than teaching orientation, international collaboration, being male). However, survey data could not be linked to publication and journal metadata, so the link between top performers and top journals could not be established (Kwiek, 2018a). In a recent study, we found that the role of top performers in a national science system in the past three decades was unchangeable: the top 1% of Polish scientists were responsible for about 10% and the top 10% of scientists for about 50% of national knowledge production (Kwiek & Roszka, 2024). Top performers emerge from the data as builders of the structure of the scientific field.

Heckman and Moktan (2020) present an institutional argument: under conditions of oversupply of publications and competition for scarce academic positions, the system simplifies the selection to publications located in a tiny set of channels (“top-five” journals) that become formal and informal career criteria. The idea of career gatekeeping through selective access to prestigious channels allows interpretations of empirical difference between top performers and the rest of the scientific population as structural and qualitative, rather than merely quantitative. In this approach, access to top journals is an elite-reproducing mechanism: it increases visibility, enables the accumulation of citations, and strengthens reputation.

Recently, several strands of research have provided additional arguments to strengthen theoretical theses regarding the increasing importance of top journals. Zhang and Yu (2020) analyzed the publishing strategies of 2,982 scientists and determined that publishing in prestigious journals early in one’s career is treated as a key quality signal. Hou et al. (2022) studied 1,933 business scholars and showed that early publications in top journals are powerful predictors of long-term career success. Additionally, early bloomers in science are more likely to attract top scientists as collaborators or have opportunities in prestigious institutions. Kraus and colleagues (2023), analyzing the career factors of 100 prominent researchers, highlight the importance of early career factors for future impact, which is consistent with the intuition that entering prestigious channels early can change career trajectories. They reported that 95% of the prominent researchers studied had at least one of four features in the first five years of their careers: working at a top-25 ranked university, publishing a paper in a top-five ranked journal, publishing most papers in top quartile (high impact) journals, or co-authoring with other researchers prominent in their field.

There is ever more work analyzing institutional pressures and the systemic effects of the increasing role of top journals. Seeber (2024) showed that changes in the publishing business may have an impact on authors’ decisions on where to publish and their focus on a narrower pool of journal titles. He observed that individual choices regarding where to publish are relevant not only for individual careers and recognition – they also collectively shape scientific communication. Trueblood and colleagues (2025) described a “misalignment of incentives” in academic publishing suggesting that emphasis on publishing in prestigious journals may generate undesirable behaviors among academics, constrain their innovativeness, and strengthen their research conformism.